

Rupesh Devasam

+1 (667) 419-0966 | rupeshdevasam@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Maryland, USA

SUMMARY

AI Full-Stack Engineer (**M.S. Computer Science, UMBC – May 2026**) with **5+ years** building production GenAI and full-stack systems for enterprise and Fortune 500 clients. Specialized in **Azure AI Agents, Prompt Flow, RAG, FastAPI**, and **React/Next.js**. Currently shipping **FDA regulatory automation (CHAAP)** and **budget intelligence (BIS)** platforms at Centennial Technologies. Strong in Python, TypeScript, distributed REST APIs, and cloud-native architecture on **Azure and AWS**.

SKILLS

- **Languages:** Python, JavaScript (ES6+), TypeScript, Java, C++, C, SQL, Shell Scripting
- **Cloud:** Azure (AI Agents, Prompt Flow, OpenAI, Web Apps, CosmosDB), AWS (Lambda, S3, EC2, RDS, DynamoDB)
- **Frameworks/Libraries:** FastAPI, Flask, Node.js, Express.js, React.js, Next.js, Vue.js, Three.js, Tailwind CSS, GraphQL, WebSockets
- **Databases:** PostgreSQL, CosmosDB, MongoDB, MySQL, Redis, DynamoDB
- **DevOps:** Docker, CI/CD, Git, GitHub Actions, Azure DevOps, RESTful API design, Microservices
- **AI/ML:** Azure AI Agents, Azure Prompt Flow, Azure OpenAI, Semantic Kernel, LangChain, MCP (Model Context Protocol), RAG, Prompt Engineering, LLM Fine-tuning, n8n Workflows, NLP, Computer Vision, PyTorch, TensorFlow, PySyft, Federated Learning, CNNs
- **Others:** Postman, Figma, Jira, Confluence, Agile/Scrum, TDD, Code Reviews, System Design

EXPERIENCE

AI Full-Stack Engineer | Centennial Technologies, Virginia, USA **Feb 2026 – Present**

- **Architected** enterprise GenAI platform powering **CHAAP** (FDA regulatory automation) and **BIS** (Budget Intelligence Suite) using **Azure AI Agents, Prompt Flow, and Azure OpenAI**, processing **15,000+ documents/month** with **72% reduction in manual review time** and **sub-3s analysis latency**.
- Developed backend services using **FastAPI and Flask** with **async processing, Redis caching, and multi-tenant data isolation** across **PostgreSQL, CosmosDB, and Redis**, achieving **180ms p95 API latency** and supporting **2,500+ concurrent users**.
- Implemented **n8n automation pipelines** orchestrating **120+ workflows** across upstream data sources, AI services, and downstream business systems, reducing manual operational work by **65%** and saving **40+ hours/week** of engineering and ops effort.
- Led frontend modernization with **React, Next.js, and TypeScript** and deployed production workloads on **Azure Web Apps** via CI/CD, reducing **page load times by 38%** and cutting **deployment time from 45 minutes to 8 minutes** with zero-downtime releases.

Full-Stack Developer Intern | CloudLeap Technologies, Baltimore, MD **Sep 2025 – Dec 2025**

- Built immersive **3D web applications** using **React.js and Three.js**, delivering browser-native interactive visualizations and gamified user experiences supporting **8,000+ active users** at **60 FPS** on standard hardware.
- Developed **Flask RESTful APIs** and drove **frontend performance optimization** via **lazy loading, memoization, and code splitting**, reducing **time-to-interactive by 42%** and **bundle size by 35%**; contributed to sprint planning, code reviews, and full-stack integration.

Software Development Engineer Intern | Centennial Technologies, Virginia, USA **Jun 2025 – Aug 2025**

- Spearheaded development of **CHAAP**, an FDA regulatory support application, building it from the ground up using **React.js (TypeScript), Python Flask, and Azure AI Services**; partnered with PMs on system architecture and CI/CD, **reducing release cycle time by 30%**.
- Awarded **“Sprint Champion”** for consistently delivering high-impact features and ensuring smooth full-stack integration under tight deadlines.

Software Engineer II | SurveyMonkey, Bangalore, India **Mar 2024 – Jun 2024**

- Developed full-stack features for high-traffic survey applications using **React, FastAPI, and Python**, supporting **100K+ monthly active users** with improved performance and scalability.
- Built reusable **React component libraries** integrated with FastAPI backends, **cutting development time by 30%**; optimized REST APIs via query optimization and caching, **reducing response times by 25%**.

Full-Stack Developer | ValueLabs, Hyderabad, India **Aug 2019 – Feb 2024**

- Developed and maintained scalable enterprise web applications for **Fortune 500 clients** using **React, Vue.js, Node.js, and Express.js**; integrated SQL and MongoDB with indexing and connection pooling, **improving data processing speed by 25%**.
- Designed and improved **RESTful APIs in Node.js/Express**, **reducing response times by 30%** through query optimization; recognized with the **Star Award** for technical contributions and mentoring junior developers.

EDUCATION

- **Master of Science in Computer Science** | University of Maryland, Baltimore County (UMBC), USA, **May 2026**
- **Bachelor of Science in Computer Science** | JNTU Kakinada (AITAM), India, **Sep 2020**